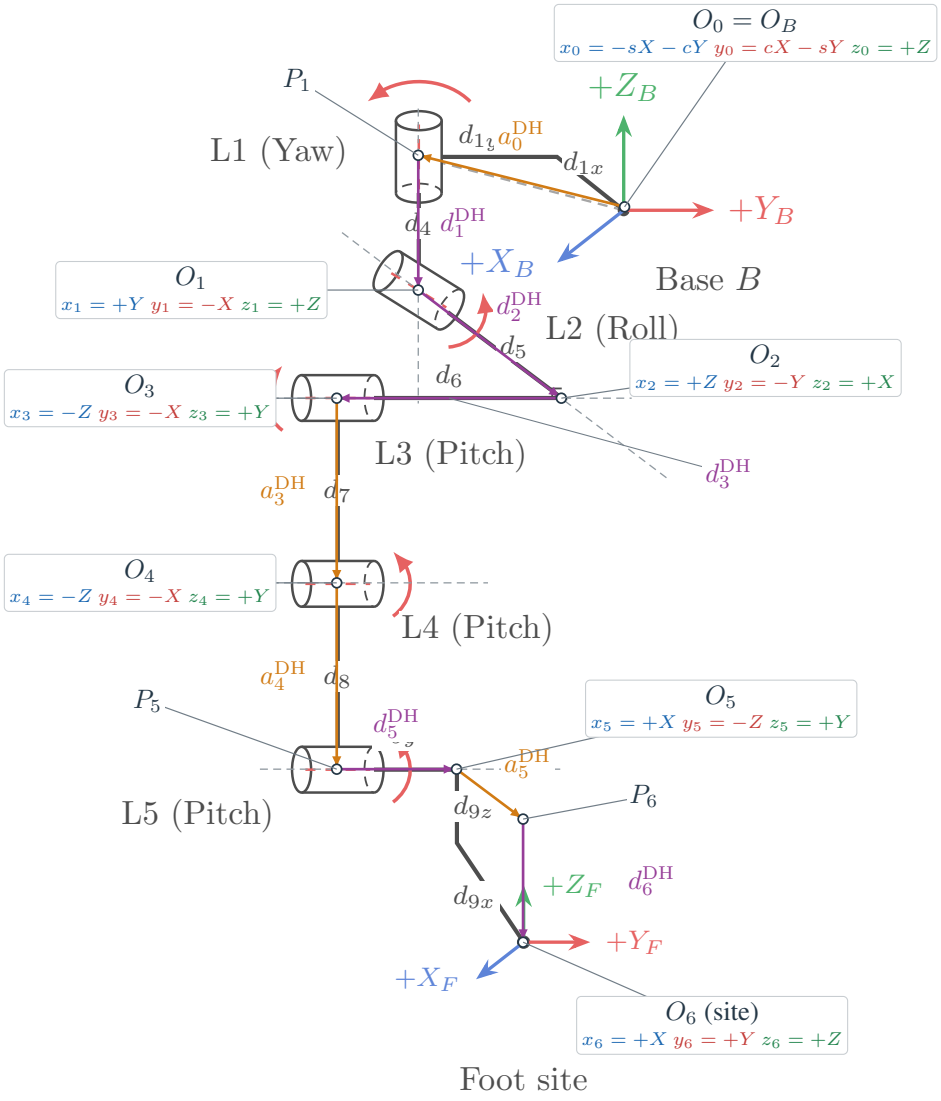
Right leg: origins, D-H distances and axes

Direct annotations on the original assignment drawing. Zero pose; distances in mm.



Orange: a_{i-1}^{DH} Purple: d_i^{DH}

Original black d labels are handout dimensions. Coloured superscript DH labels are table parameters.

Read each row as a two-step path

$$O_{i-1} \xrightarrow[+x_{i-1}]{a_{i-1}} P_i \xrightarrow[+z_i]{d_i} O_i$$

A negative value means travel opposite to that positive axis.

i	a_{i-1}	d_i	axis for a	axis for d
1	39.8246	−0.0741	$-sX - cY$	$+Z$
2	0	−16.06	$+Y$	$+X$
3	0	−74.15	$+Z$	$+Y$
4	78.65	0	$-Z$	$+Y$
5	78.65	17.35	$-Z$	$+Y$
6	0.5	−36.225	$+X$	$+Z$

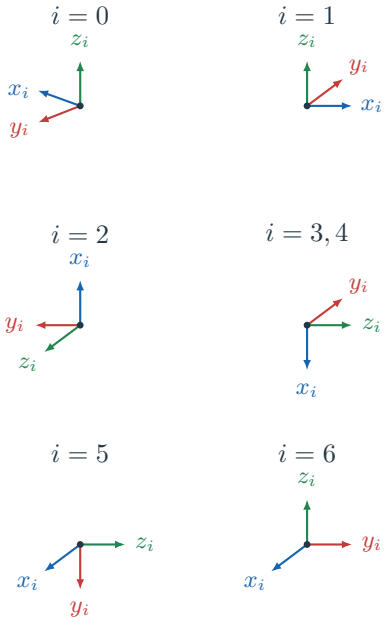
$a_1 = a_2 = 0$; $d_4 = 0$.
$P_2 = O_1$, $P_3 = O_2$, $P_4 = O_4$.

Full frame axes, expressed in base B

frame	$+x_i$	$+y_i$	$+z_i$
0	$-sX - cY$	$cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6	$+X$	$+Y$	$+Z$

$s = \sin\beta$, $c = \cos\beta$, $\beta = \text{atan2}(19, 35)$.

Positive-axis triads
Translated from their O_i



O_1: yaw/roll axis intersection. O_2: roll/hip-pitch axis intersection. P_5 and O_5 share the ankle axis. P_6 is the auxiliary corner for the D-H translation order. $O_0 = B$. The first vertical segment ends at O_1; $d_1 = -0.0741$ mm. Cylinder centres are not assumed to be D-H origins.

Original: Assignment 1, Fig. 3. Schematic, not to scale; perspective is illustrative. Exact signs and directions are given in the tables. Markers are selected D-H origins, not CAD joint centres.